

KAREN L. SMITH – CURRICULUM VITAE

Dept. of Physical and Environmental Sciences
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A. EDUCATION

2011	Ph.D., Physics, University of Toronto, Toronto, Canada
2007	M.Sc., Nutritional Sciences, University of Toronto, Toronto, Canada
2004	M.Sc., Environmental Science and Engineering, California Institute of Technology, Pasadena, CA, USA
2002	B.Sc. (1 st Class), Mathematics and Engineering, Queen's University, Kingston, Canada

B. PROFESSIONAL EXPERIENCE

University of Toronto Scarborough, Toronto, Canada

2023 – present	Associate Professor, Teaching Stream • Director, Master of Environmental Science Program (MEnvSc), 2025-present • Program Director, Climate Change Impacts and Adaptation MEnvSc Field-of-Study, 2017–present
2017 – 2023	Assistant Professor, Teaching Stream

Lamont-Doherty Earth Observatory of Columbia University, Palisades, NY, USA

2023 – present	Adjunct Research Scientist
2017– 2023	Adjunct Associate Research Scientist
2014 – 2017	Associate Research Scientist • Visiting Scientist, NASA Goddard Institute for Space Studies, New York, NY, USA, 2015–2017
2013 – 2015	Natural Sciences and Engineering Research Council of Canada Postdoctoral Fellow
2011 – 2013	Postdoctoral Research Scientist

C. GRANTS, AWARDS AND CERTIFICATES

1. Grants

1.1 Current

16. “Enhancing climate literacy among environmental science professionals”, PI: Karen Smith; UTSC CTL Teaching Enhancement Grant, \$3,000.
15. “Bridging Environments: Engaging Indigenous Perspectives in Environmental Sciences”, PI: Stuart Livingstone; Co-PIs: Karen Smith, Élyse Caron-Beaudoin, Cody Ross; UTSC Experiential Learning Fund 02/07/2025 to 04/30/2026; \$11,500.

1.2 Previously Funded

14. *"The impacts of climate change on major resource projects and associated environmental, health, social and economic effects"*, PI: Karen Smith, Impact Assessment Agency of Canada Research Program 01/01/2024 to 12/31/2024; \$72,000.
13. *"Rapid Adjustments and Their Effects on Arctic Amplification"*, PI: Michael Previdi (LDEO); Co-PIs: Lorenzo Polvani (LDEO/Columbia), Karen Smith (UTSC), National Science Foundation (USA) PLR-2234876, from 03/01/2023 to 02/28/2026; \$760,789 USD.
12. *"Application of the University of Toronto Climate Downscaling Workflow to the Just Energy Transition"*, PI: Paul Kushner (UTSG); Co-PIs: Sam Markolf (UC Merced), Oya Mercan (UTSG), Daniel Posen (UTSG), Karen Smith (UTSC), Marianne Touchie (UTSG), University of Toronto Climate Positive Energy Just Energy Transition Grant from 01/01/2023-12/31/2024; \$43,000.
11. *"Unlearning Environment: Engaging Indigenous Perspectives in Environmental Sciences"*, PI: Stuart Livingstone (UTSC); Co-PIs: Karen Smith (UTSC), Taylor Tabobondung (UTSC), Élyse Caron-Beaudoin (UTSC), UTSC Pedagogies of Inclusive Excellence Circle 1 Grant from 05/01/2023 to 04/30/2024; \$18,000.
10. *"Collaborative Case-based Experiential Learning: Online Platform Development"*, PI: James MacLellan; Co-PIs: Karen Smith, Laura Tozer, Nicole Klenk, UTSC Experiential Learning Fund from 06/01/2023 to 05/31/2024; \$8,640.
9. *"A Canadian Meteorological and Oceanographic Society (CMOS) Initiative in partnership with Canadian Water Resources Association (CWRA) and the Black Environmental Initiative (BEI) to address Diversity and Inclusiveness in the meteorological, climatological, and hydrological science communities in Canada"*, PI: James Abraham (rep. CMOS); Co-PIs: Naolo Charles (BEI), Russel Boals (rep. CWRA), Karen Smith (rep. CMOS), from 09/01/2022-03/31/2025; \$175,000.
8. *"A pilot micro-credential in the DPES MEnvSc Program: Indigenous perspectives in environmental sciences"* PI: Karen Smith (UTSC); Co-PIs: Stuart Livingstone (UTSC), Élyse Caron-Beaudoin (UTSC), David McLagan (Queen's); Collaborator: Taylor Tabobondung (UTSC), University of Toronto Graduate Education Innovation Fund from 05/13/2022-08/31/2023; \$5,000.
7. *"Climate change, extreme heat, and human mortality in the City of Toronto"*, PI: Rohan Alexander (UTSG); Co-PIs: Monica Alexander (UTSG); Karen Smith (UTSC); William Gough (UTSC); Collaborators: Samantha Green (UTSG); Edward Xie (UTSG); Laura Tozer (UTSC); Micah Hewer (UTSC), School of Cities Urban Challenge Grant 2.0 from 01/01/2022-04/30/2023; \$75,000.
6. *"Earth and Environmental Data Science Immersive Workshop"*, PI: Amir Feizpour (Aggregate Intellect); Co-PIs: Karen Smith (UTSC) and Andre Erler (Aquanty Inc.); Natural Resources Canada 2021-2022 Science Outreach and Engagement Grant (theme: "Hackathons or other data science challenges to help unlock the value of data for innovation and sustainable development in Canada's natural resource sectors") from 01/01/2022 to 06/30/2022; \$37,000.
5. *"Engineering in a Changing Climate - A Transdisciplinary Workshop Series for Engineering and Climate Science Students"*, PIs: Karen Smith (UTSC) and Oya Mercan (UTSG), Co-PIs: Marianne Hatzopoulou (UTSG), Paul Kushner (UTSG), Reza Najafi (Western), Graeme Norval (UTSG), Daniel Posen (UTSG), Marianne Touchie (UTSG), eCampus Ontario Virtual Learning Strategy, from 04/15/2021 to 03/31/2022; \$39,456.45.
4. *"Building and Governing Climate Resilient Cities"*, PI: Fadi Masoud (UTSG); Co-PIs: Amy Buitenhuis (City of Toronto), Fariha Husain (City of Toronto), Jia Lu (City of Toronto), Paul

Kushner (UTSG), Heather Marshall (Toronto Environmental Alliance), Oya Mercan (UTSG), Blake Poland (UTSG), Rehana Rajabali (TRCA), Imara Rolston (UTSG), John Robinson (UTSG), Karen Smith (UTSC), Linda Swanston (City of Toronto), University of Toronto School of Cities Urban Challenge Group (Stage 1 Approved, February 2019; Stage 2 Approved, March 2021).

3. *"Climate Science for Engineering Decision, Education and Policy (CSE)"*, Project Lead: Oya Mercan (UTSG), Team: Marianne Hatzopoulou (UTSG), Paul Kushner (UTSG), Graeme Norval (UTSG), Daniel Posen (UTSG), Karen Smith (UTSC), Marianne Touchie (UTSG). University of Toronto, Faculty of Applied Science, Dean's Strategic Fund, DSF18-30, from 02/01/2019 to 06/30/2023; \$384,000.
2. *"Enhancing Climate Change Science Education at UTSC using Authentic, Model-based Inquiry"*, PI: Karen L. Smith (UTSC); Co-PI: Tanzina Mohsin (UTSC), UTSC Teaching Software Grant, Winter 2018, \$2,647.30.
1. *"The Impact of the Stratosphere on Arctic Climate"*, PI: Karen L. Smith (LDEO); Co-PIs: Lorenzo M. Polvani (LDEO/Columbia), Bruno Tremblay (McGill), Michael Previdi (LDEO); Collaborator: Douglas E. Kinnison (NCAR), National Science Foundation (USA) PLR-1603350, from 07/15/2016 to 12/31/2021; \$600,795 USD.

2. Awards

2.1 Professional Awards

2023	Neil J. Campbell Award for Exceptional Volunteer Service to the Canadian Meteorological and Oceanographic Society
2023	American Geophysical Union Voices for Science Award
2022	UTSC Assistant Professors and Lecturers Teaching Award

2.2 Postdoctoral and Student Fellowships and Awards

2016	Insight Data Science Fellowship, New York, NY, USA
2013	Natural Sciences and Engineering Research Council of Canada (NSERC) Postdoctoral Fellowship
2009	Marie Curie Sklodowska Association Award, University of Toronto
2009	Seymour H. Vosko Memorial Prize, University of Toronto
2009	Canadian Meteorological and Oceanographic Society Best Student Poster Prize
2007	NSERC Postgraduate Scholarship (PGS), Doctoral
2006	Max and Ruth Wiseman Graduate Student Fellowship, Baycrest Centre for Geriatric Care
2006	NSERC PGS, Master's
2003	Bill Davidow Graduate Student Fellowship, California Institute of Technology
2002	Vito Vanoni Fellowship, California Institute of Technology
2002	Special Institute Fellowship, California Institute of Technology
2002	Annie Bentley Lillie Prize in Mathematics, Queen's University

3. Certificates

2019 – 2020	Centre for Teaching and Learning (CTSI) - Association for College and University Educators (ACUE) Certificate in Effective Teaching Practice Certificate
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D. TEACHING EXPERIENCE

1. Graduate Courses Taught

I developed the syllabus and all lectures, labs and assessments for the following University of Toronto graduate courses, for which I have sole responsibility (except for EES1100 which was co-taught). For information on Readings courses supervised, see Section E.

2017 – present	EES1133: Climate Change Science and Modeling , 8-25 students, 2 h/w lecture, 1h/w computer lab, Fall 2017, 2018, 2019, 2020 (virtual), 2021 (virtual), 2022, 2023, 2025
2017 – present	EES1132 (cross-listed with EESD21): Geophysical and Climate Data Analysis , 8-21 students, 3 h/w lecture, Fall 2017, 2018, 2019, 2020 (virtual), 2021 (virtual), 2022 (not offered), 2023, 2025
2017 – present	EES1117: Climate Change Impact Assessment , 8-22 students, 2 h/w lecture, 1 h/w computer lab, Winter 2018, 2019, 2020, 2021 (virtual), 2022 (hybrid), 2023, 2024, 2026
2017 – 2022	EES1100: Advanced Topics in Environmental Science , 85-100 students, co-taught (three instructors), 1.5 h/w, every other week, Fall-Winter 2017-2018, 2018-2019, 2019-2020, 2020-2021 (virtual), 2021-2022 (virtual)

2. Undergraduate Courses Taught

2018 – 2024	EESC24: Advanced Readings in Environmental Science (Supervisor Only), 1-7 students, Fall-Winter-Summer, 2018-2019, 2019-2020, 2020-2021, 2021-2022, Summer 2023, Fall-Winter 2023-2024
2023 – present	EESD21 (cross-listed with EES1132): Geophysical and Climate Data Analysis , 8-21 students, 3 h/w lecture, Fall 2023, 2025

3. Co-curricular Workshops/Events Offered

2024	U of T Climate Impacts Hackathon , University of Toronto
2023	Indigenous Perspectives in the Environmental Sciences (two offerings, one in-person and one online), University of Toronto Scarborough
2022	Earth and Environmental Data Science Immersive Workshop, Virtual (hosted by Aggregate Intellect)
2018 – present	Graduate Professional Development Workshop: Basic and Intermediate Python, University of Toronto Scarborough
2020	Centre for Climate Science and Engineering Workshop: Global Climate Modeling and Regional Downscaling, University of Toronto

4. Other

2012 – 2016	Course Scientist, Seminars on Science American Museum of Natural History, New York, NY
2008 – 2011	Teaching Assistant, Patterns from Chaos Dept. of Physics, University of Toronto, Toronto, Canada
2006	Teaching Assistant, Advanced Nutritional Sciences Dept. of Nutritional Sciences, University of Toronto, Toronto, Canada

2004 Teaching Assistant, Atmosphere and Ocean Dynamics
Dept. of Geological and Planetary Sciences, California Institute of Technology,
Pasadena, CA

E. PUBLICATIONS

04/24/2026: 34 Peer-reviewed publications; [Google Scholar](#): 2778 citations; h-index = 23 (* indicates current or former graduate student)

1. Refereed; Published or In Press

34. Morris, M., P. J. Kushner, **K. L. Smith**, The University of Toronto Climate Downscaling Workflow Tools and Resources for Climate Change Impact Analysis, *Journal of Open Source Education*, 7(78), 243, doi:10.21105/jose.00243.
33. Bu, S.*, **K. L. Smith**, F. Masoud, A. Sheinbaum, 2024: Spatial distribution of heat vulnerability in Toronto, *Urban Climate*, 54, doi: 10.1016/j.uclim.2024.101838.
32. Liang, Y-C., Y.-O. Kwon, C. Frankignoul, G. Gastineau, **K. L. Smith**, L. M. Polvani, L. Sun, Y. Peings, C. Deser, R. Zhang, and J. Screen, 2024: The weakening of the stratospheric polar vortex and the subsequent surface impacts as consequences to Arctic sea-ice loss, *Journal of Climate*, 37, 309-333, doi: 10.1175/JCLI-D-23-0128.
31. Janoski, T. P.*, M. Previdi, G. Chiodo, **K. L. Smith**, and L. M. Polvani, 2023: Ultrafast Arctic amplification and its governing mechanisms, *Environmental Research: Climate*, doi: 10.1088/2752-5295/ace211.
30. Liang, Y-C., L. M. Polvani, M. Previdi, **K. L. Smith**, M. R. England and G. Chiodo, 2022: Stronger Arctic amplification from ozone-depleting substances than from carbon dioxide, *Environmental Research Letters*, 17, 024010, doi: 10.1088/1748-9326/ac4a31.
29. Anderson, C. and **K. L. Smith**, 2021: A narrative approach to building computational capacity in professional Master's students, *Journal of Open Source Education*, doi: 10.21105/jose.00100.
28. Previdi, M., **K. L. Smith**, and L. M. Polvani, 2021: Arctic amplification of climate change: a review of underlying mechanisms, *Environmental Research Letters*, 16, 093003, doi: 10.1088/1748-9326/ac1c29.
27. **Smith, K. L.** and L. M. Polvani, 2021: Modeling-evidence for large, ENSO-driven interannual wintertime AMOC variability, *Environmental Research Letters*, 16, 084038, doi: 10.1088/1748-9326/ac1375.
26. Previdi, M., T. P. Janoski*, G. Chiodo, **K. L. Smith** and L. M. Polvani, 2020: Arctic amplification as a rapid adjustment to increased CO₂, *Geophysical Research Letters*, 47, e2020GL089933, doi: 10.1029/2020GL089933.
25. Maleska, S.* **K. L. Smith** and J. Virgin*, 2020: Impacts of stratospheric ozone extremes on Arctic high cloud, *Journal of Climate*, doi: 10.1175/JCLI-D-19-0867.1

24. Polvani, L. M, M. Previdi, M. England, G. Chiodo and **K. L. Smith**, 2020: Large Arctic warming from ozone depleting substances in the latter half of the 20th Century. *Nature Climate Change*, 10, 130–133, doi: 10.1038/s41558-019-0677-4.
23. Virgin, J.* and **K. L. Smith**, 2019: Is Arctic Amplification dominated by regional radiative forcing and feedbacks? Perspectives from the World-Avoided scenario, *Geophysical Research Letters*, 46, 7708–7717, doi:10.1029/2019GL082320.
22. Zhang, P., Y Wu, I Simpson, **K. L. Smith**, B. De, P. Callaghan, 2018: A stratospheric pathway linking a colder Siberia to Barents-Kara sea ice loss, *Science Advances*, 4, 7, eaat6025, doi: 10.1126/sciadv.aat6025.
21. **Smith, K. L.**, L. M. Polvani and L. Bruno Tremblay, 2018: The impact of stratospheric circulation extremes on minimum Arctic sea ice extent, *Journal of Climate*, 31, 7169–7189, doi:10.1175/JCLI-D-17-0495.1.
20. **Smith, K. L.**, G. Chiodo, M. Previdi and L. M. Polvani, 2018: No surface cooling over Antarctica from the negative greenhouse effect associated with instantaneous quadrupling of CO₂ concentrations, *Journal of Climate*, 31, 317–323, doi: 10.1175/JCLI-D-17-0418.1.
19. Zhang, P., Y. Wu and **K. L. Smith**, 2017: Prolonged effect of the stratospheric pathway in linking Barents-Kara Sea sea ice variability to the midlatitude circulation in a simplified model, *Climate Dynamics*, doi: 10.1007/s00382-017-3624-y.
18. **Smith, K. L.** and L. M. Polvani, 2017: Spatial patterns of recent Antarctic surface temperature trends and the importance of natural variability: Lessons from multiple reconstructions and the CMIP5 models, *Climate Dynamics*, doi: 10.1007/s00382-016-3230-4.
17. England, M. R., L. M. Polvani, **K. L. Smith**, L. Landrum and M. M. Holland, 2016: Robust response of the Amundsen Sea Low to stratospheric ozone depletion, *Geophysical Research Letters*, doi: 10.1002/2016GL070055.
16. Wu, Y. and **K. L. Smith**, 2016: Response of the Northern Hemisphere midlatitude circulation to Arctic amplification in a simple atmospheric general circulation model, *Journal of Climate*, doi: 10.1175/JCLI-D-15-0602.1.
15. **Smith, K. L.** and R. K. Scott, 2016: The role of planetary waves in the tropospheric jet response to stratospheric cooling, *Geophysical Research Letters*, doi: 10.1002/2016GL067849.
14. Previdi, M., **K. L. Smith** and L. M. Polvani, 2015: How well do the CMIP5 models simulate the Antarctic Atmospheric Energy Budget, *Journal of Climate*, doi.org/10.1175/JCLI-D-15-0027.1.
13. Solomon, A., L. M. Polvani, R. Abernathy and **K. L. Smith**, 2015: The ozone hole's effects on the state of the Southern Ocean, *Geophysical Research Letters*, doi.org/10.1002/2015GL064744.
12. Neely, R. R., D. R. Marsh, **K. L. Smith**, S. M. Davis and L. M. Polvani, 2014: Biases in Southern Hemisphere climate trends induced by coarsely specifying the temporal resolution of stratospheric ozone, *Geophysical Research Letters*, doi: 10.1002/2014GL061627.

11. **Smith, K. L.**, R. R. Neely, D. R. Marsh and L. M. Polvani, 2014: The Specified Chemistry Whole Atmosphere Community Climate Model (SC-WACCM). *Journal of Advances in Modeling the Earth System*, doi: 10.1002/2014MS000346.
10. **Smith, K. L.** and L. M. Polvani, 2014: The surface impacts of Arctic stratospheric ozone anomalies. *Environmental Research Letters*, 9, 074015.
9. **Smith, K. L.**, M. Previdi and L. M. Polvani, 2013: The Antarctic atmospheric energy budget. Part II: The effect of ozone depletion and its projected recovery. *Journal of Climate*, Vol. 26, 9729-9744.
8. Previdi, M., **K. L. Smith** and L. M. Polvani, 2013: The Antarctic atmospheric energy budget. Part I: Climatology and intraseasonal-to-interannual variability. *Journal of Climate*, 26, 6406-6418.
7. Polvani, L. M. and **K. L. Smith**, 2013: Can natural variability explain the observed sea ice trends? New modeling evidence from CMIP5. *Geophysical Research Letters*, 40, 12, 3195–3199.
6. **Smith, K. L.**, L. M. Polvani and D. R. Marsh, 2012: Mitigation of 21st century Antarctic sea ice loss by stratospheric ozone recovery. *Geophysical Research Letters*, 39, 20, L20701.
5. **Smith, K. L.** and P. J. Kushner, 2012: Linear interference and the initiation of extratropical stratosphere-troposphere interactions. *Journal of Geophysical Research*, 117, D13107, doi.org/10.1029/2012JD017587.
4. **Smith, K. L.**, P. J. Kushner, and J. Cohen, 2011: The role of linear interference in Northern Annular Mode variability associated with Eurasian snow cover extent. *Journal of Climate*, 24, 6185-6202.
3. **Smith, K. L.**, C. G. Fletcher, and P. J. Kushner, 2010: The role of linear interference in the Annular Mode response to extratropical surface forcing. *Journal of Climate*, 23, 6036-6050.
2. **Smith, K. L.** and C. E. Greenwood, 2008: Nutritional considerations and Alzheimer's disease. *Journal of Nutrition for the Elderly*, 27, 3, 381-403.
1. Schneider, T., **K. L. Smith**, P. A., O'Gorman, and C. C., Walker, 2006: A climatology of tropospheric zonal-mean water vapor fields and fluxes in isentropic coordinates. *Journal of Climate*, 19, 5918-5933.

2. Non-refereed Publications

7. **Smith, K. L.**, N. Feldl, R. Caballero, P. Keys, 2025: Focus on Arctic Amplification. *Journal of Environmental Research: Climate*. Editorial for Special Collection on Arctic Amplification.
6. **Smith, K. L.**, Cassidy, C., Abraham, J., Charles, N., Goodman, A. (rep.). Canadian Meteorological and Oceanographic Society (CMOS), 2023: Report on Equity, Diversity, Inclusion and Accessibility (EDIA) Initiative.
5. Garny, H., H. Hendon; Co-Authors: M. Abalos, G. Chiodo, A. Purich, W. Randel, **K. Smith** and D. Thompson: Stratospheric Ozone Changes and Climate, Chapter 5 in World Meteorological Organization (WMO). Scientific Assessment of Ozone Depletion: 2022, GAW Report No. 278, 509 pp.; WMO: Geneva, 2022.

4. **Smith, K. L.** [The unexpected link between the ozone hole and Arctic warming](#), The Conversation, February 2020.
3. Kushner, P. J., **K. L. Smith**, R. D. Brown, C. Dersken, C. R. Duguay, R. Fernandes, and W. R. Peltier. Workshop report: Simulation of the Canadian cryosphere. *CMOS Bulletin*, 38, 2, 2010, 60-65.
2. **Smith, K. L.**, C. E. Greenwood, H. Payette, and S. M. H. Alibhai, 2007: An approach to the non-pharmacologic and pharmacologic management of unintentional weight loss among older adults. *Geriatrics and Aging*, Vol. 10, No. 2, 91-98.
1. **Smith, K. L.**, C. E. Greenwood, H. Payette, and S. M. H. Alibhai, 2006: An approach to the diagnosis of unintentional weight loss in older adults: prevalence rates and screening. *Geriatrics and Aging*, 9, 10, 679-685.

3. Creative Works

1. Livingstone, S. and **K. L. Smith**, [Emerging Environments Podcast](#), Seasons 1, 2021 (13 episodes + 2 bonus episodes), Season 2, 2022 (12 episodes), Season 3, 2023 (8 episodes), Season 4, 2024 (8 episodes), Season 5, 2025 (7 episodes).

F. INVITED INSTITUTIONAL AND CONFERENCE PRESENTATIONS

2026	University of Waterloo, Department of Applied Mathematics, Waterloo, Canada
2024	McGill University, Department of Atmospheric and Oceanic Sciences, Montréal, Canada
2023	Extreme Heat Emergencies Planning Session, Toronto Emergency Management, Toronto, Canada
2023	Occupational and Environmental Health Seminar, Dalla Lana School of Public Health, University of Toronto, Toronto, Canada
2021	European Geosciences Union Conference (Session: The role of the middle atmosphere: composition changes and feedbacks, Vienna, Austria (virtual))
2021	University of Toronto, Centre for Global Change Science, Toronto, Canada (virtual)
2019, 2023	Royal Conservatory of Music, Toronto, Canada
2018	Université de Québec à Montréal, ESCER, Montréal, Canada
2017	Aspen Global Change Institute, Aspen, CO, USA
2016	American Geophysical Union (Session: The dynamical processes of stratosphere-troposphere interactions and their relationship with climate) , San Francisco, CA, USA
2016	University of Toronto, Department of Physics, Toronto, Canada
2016	University of Toronto Scarborough, DPES, Toronto, Canada
2016	University of Waterloo, Department of Applied Mathematics, Waterloo, Canada
2015	NASA Goddard Institute for Space Studies, New York, NY, USA
2015	Stony Brook University, School of Marine and Atmospheric Sciences, Stony Brook, NY

2015	University of California, Berkeley, Department of Earth and Planetary Science, Berkeley, CA, USA
2015	Lamont-Doherty Earth Observatory, Palisades, NY, USA
2014	University of Toronto, Department of Physics, Toronto, Canada
2014	York University, Department of Earth and Space Science and Engineering, Toronto, Canada
2013	NASA Goddard Institute for Space Studies, New York, NY, USA
2013	Dalhousie University, Department of Physics and Atmospheric Science, Halifax, Canada
2013	MIT, Department of Earth, Atmospheric and Planetary Sciences, Boston, MA
2013	McGill University, Department of Atmospheric and Oceanic Sciences, Montreal, Canada
2013	Pennsylvania State University, Department of Meteorology, State College, PA
2012	Johns Hopkins University, Department of Earth and Planetary Sciences, Baltimore, MD
2012	New York University, Courant Institute, New York, NY, USA
2011	Lamont-Doherty Earth Observatory, Palisades, NY, USA
2010	United Kingdom Meteorological Office, Exeter, UK
2010	University of Reading, Department of Meteorology, Reading, UK
2009	A.E.R., Inc., Lexington, MA, USA

G. SERVICE AND OUTREACH

1. Institutional Service

2025 – present	Director of MEnvSc Program, DPES, UTSC
2017 – present	Program Director of Climate Change Impacts and Adaptation MEnvSc Program, DPES, UTSC
2017 – present	DPES Graduate Program and Curriculum Committee
2018 – present	DPES MEnvSc Admissions Committee
2018 – 2024	DPES Teaching and Curriculum Committee
2018 – 2024	DPES Outreach Committee
2021 – present	UTSC Campus Curriculum Committee
2020 – 2022	UTSC Library Advisory Committee
2021	DPES PTR Committee

2. Professional Service

2.1 Scientific Assessments

2023 – present	Coordinating Lead Author, Chapter 4: Large-scale Atmosphere-Ocean Processes, Canada's Changing Climate Report 2025
2021 – 2022	Lead Author, Chapter 5: Stratospheric ozone changes and climate, World Meteorological Organization Scientific Assessment of Ozone Depletion 2022

2.2 Organizations and Committees

2020 – present	Chair, Canadian Meteorological and Oceanographic Society (CMOS) School and Public Education Committee
2022 – present	Member, CMOS University and Professional Education Committee
2019 – 2023	Executive Team Member, Centre for Climate Science and Engineering, University of Toronto
2019 – 2020	Member, CMOS School and Public Education Committee
2019 – 2022	Member, Canadian Centre for Climate Services, Training Sub-committee

2.3 Editor, Peer-reviewer

Editor: [Journal of Open Source Education \(JOSE\)](#)

Guest Editor: [Focus Collection on Arctic Amplification, Environmental Research: Climate](#)

Reviewer, Journals: Science, Nature Climate Change, Nature Communications, Journal of Climate, Journal of the Atmospheric Sciences, Journal of Geophysical Research (Atmospheres), Geophysical Research Letters*, Climate Dynamics, Environmental Research Letters, JOSE, Weather and Climate Dynamics, EGU-Sphere, Atmospheric Chemistry and Physics, Ecological Informatics

**Received 2019 Editors' Citation for Excellence in Refereeing*

Reviewer, Funding Agencies: Natural Sciences and Engineering Research Council (Canada), National Science Foundation (USA), Israel Science Foundation (Israel), Marsden Foundation (New Zealand), New Zealand Antarctic Research Institute, Swiss Federal Institute of Technology

Reviewer, Scientific Assessments: World Meteorological Organization Scientific Assessment of Ozone Depletion 2014, 2018

Reviewer, Other: Toronto and Region Conservation Authority, Partners in Project Green, Building Business Resilience to Climate Change – Business Guide

H. PROFESSIONAL ASSOCIATIONS

2008 – present	American Geophysical Union
2009 – present	Canadian Meteorological and Oceanographic Society
2009 – present	American Meteorological Society
2010 – present	European Geosciences Union